

Advanced (Habanero)

Q1. Solve the system: $x + y = 3$, $x^2 + y^2 = 9$

- (A) $(x, y) = (1, 2)$
(B) $(x, y) = (2, 1)$
(C) $x = 1, y = 2$ or $x = 2, y = 1$
(D) $x = -1, y = 4$ or $x = 4, y = -1$
(E) No solution

Q2. What is the intersection point of $y = x^2 - 2x + 1$ and $y = 2x - 3$?

- (A) $(1, -2)$
(B) $(1, 0)$
(C) $(2, 1)$
(D) $(-1, -4)$
(E) $(3, 3)$

Q3. Solve the system: $y = x^2$, $y = 2x - 1$

- (A) $x = 1, y = 1$
(B) $x = -1, y = 1$
(C) $x = 1, y = 1$ or $x = -1, y = 1$
(D) $x = 2, y = 4$
(E) No solution

Q4. What is the x -coordinate of the intersection point of $y = x^2 - 4x + 3$ and $y = x - 1$?

- (A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$

Q5. Solve the system: $x^2 + y^2 = 4$, $x + y = 2$

- (A) $(0, 2)$
(B) $(2, 0)$
(C) $(1, 1)$
(D) $(-1, -1)$
(E) No solution

Q6. What is the y -coordinate of the intersection point of $y = x^2 - 2$ and $y = 2x$?

- (A) $y = 0$
(B) $y = 1$
(C) $y = 2$
(D) $y = -2$
(E) $y = 4$

Q7. Solve the system: $y = x^2 + 1$, $y = -x + 2$

- (A) $(1, 2)$
(B) $(-1, 2)$
(C) $(1, 1)$
(D) $(-1, 0)$
(E) No solution

Q8. What is the intersection point of $y = x^2 - 1$ and $y = x + 1$?

- (A) $(1, 2)$
(B) $(-1, 0)$
(C) $(0, -1)$
(D) $(-2, -1)$
(E) $(2, 3)$

Open Ended Questions (FRQ)**Q9.** Solve the system: $x^2 + y^2 = 25$, $y = x - 5$. Show all work and explain your answer.**Q10.** Find the intersection points of the system: $y = x^2 - 4x + 3$, $y = x - 1$. Explain your method and show all work.**Chef's Tips**

- Read each question carefully
- Use a pencil to write your answers
- Check your work before submitting